



## Programming Fundamentals

### Assignment # 5

### BSCS & BSSE

**Assignment deadline Tuesday, 13 December 2022, 11:55 PM**

#### Submission Instructions:

- All problems must be solved by following the order and submitted in .cpp file in zip folder and also have to submit the word file (.docx) otherwise marks will be deducted. The folder name should be XXXX\_A05.zip. Where XXXX is your roll number.
- Every .cpp file must be named after the question number i.e. Q1.cpp, Q2.cpp
- Printed assignment will get you zero marks.
- This is an individual assignment.

**Plagiarism is strictly prohibited.**

#### Task-1

Write a Function that prints the day number of the year, given the date in the form month-day-year. For example, if the input is 1-1-2006, the day number is 1; if the input is 12-25-2006, the day number is 359. The program should check for a leap year. A year is a leap year if it is divisible by 4, but not divisible by 100. For example, 1992 and 2008 are divisible by 4, but not by 100. A year that is divisible by 100 is a leap year if it is also divisible by 400. For example, 1600 and 2000 are divisible by 400. However, 1800 is not a leap year because 1800 is not divisible by 400.

#### Task-2

Write a function that displays at the left margin of the screen a solid square of asterisks whose side is specified in integer parameter side. For example, if side is 4, the function displays the following:

```
****
****
****
****
```



### **Task-3**

Calculate power of a number x using a function “*PowerCalc*”. Input integer x and the power p from user. Pass x and p to PowerCalc function. Calculate and display first p powers of x on console in the PowerCalc function.

Example:

PowerCalc(5, 4) will give the following output:

5

25

125

625

### **Task-4**

An integer is said to be a perfect number if the sum of its divisors, including 1 (but not the number itself), is equal to the number. For example, 6 is a perfect number, because  $6 = 1 + 2 + 3$ . Write a function is Perfect that determines whether parameter number is a perfect number. Use this function in a program that determines and prints all the perfect numbers between 1 and 1000. Print the divisors of each perfect number to confirm that the number is indeed perfect.

- a) Write a function that takes an integer value and returns the number with its digits reversed. For example, given the number 7631, the function should return 1367.
- b) Write a function multiple that determines for a pair of integers whether the second is a multiple of the first. The function should take two integer arguments and return true if the second is a multiple of the first, false otherwise. Use this function in a program that inputs a series of pairs of integers. (Hint: Call this function within do-while loop).
- c) Write a program that has a function that takes 3 alphabet characters and find their preceding alphabets. For example if user entered a, m, y it will show b, n, z. Take input in main, pass values to function and show the output in main.
- d) Write a program that has a function “Upper\_to\_lower”. It will take 4 alphabets as parameters in uppercase and will change them to lowercase. Display the changed values in main.



## **Task-5**

Write a program that takes a `upper_bound` integer value from the user and passes it to a function named `sumOfNumbers`. Sum the odd and even numbers, respectively, from 1 to a given `upper_bound`. Also, compute the absolute difference between both of them and return it to the main function.

## **Task-6**

Write the definition of a function named `karatosDestruction` that takes as input the three numbers. The function returns `true` if the first number to the power of the second number equals the third number; otherwise, it returns `false`. Assume that the three numbers are of type `double`. If `false` is returned, prompt the user to again provide new values otherwise terminate the program.